

EC1208

Principles of Macro-Economy

Summer 2024 Examination
Paper 1 · 1.5 hours · Section A: answer 2 of 3 (40%) · Section B: answer 1 of 2 (60%)

Module	EC1208 — Principles of Macro-Economy
Session	Summer 2024
Structure	Section A: three short questions (100% each) — answer any two, together worth 40% of the total. Section B: two long questions with unevenly-weighted parts — answer any one, worth 60% of the total.
This scheme covers	All three Section A questions AND both Section B questions in full, so you can study every possible combination rather than just the ones you'd pick on the day.

A source-paper note on B1(c)(ii). The question states the government “encourages more people to work by lowering the retirement age” — but lowering the retirement age would let people retire *earlier*, reducing labour supply, the opposite of the stated intent. This is treated as a likely typo for “raising the retirement age” throughout this scheme, since that’s the only reading consistent with “encourages more people to work” and the ageing-population context given.

Section A — Short Questions

Answer 2 of 3 · 40% of total marks · each question worth 100% of its own mark allocation

Question A1

100%

Using a diagram, explain the relationship between a country's physical capital stock and GDP.

FULL MODEL ANSWER

Physical capital — the stock of business structures and equipment used in production — is one of the two key inputs (alongside efficiency units of labor) in the aggregate production function, $Y = AF(K, H)$. Holding efficiency units of labor and technology constant, the relationship between K and GDP is captured by a single upward-sloping, concave curve.

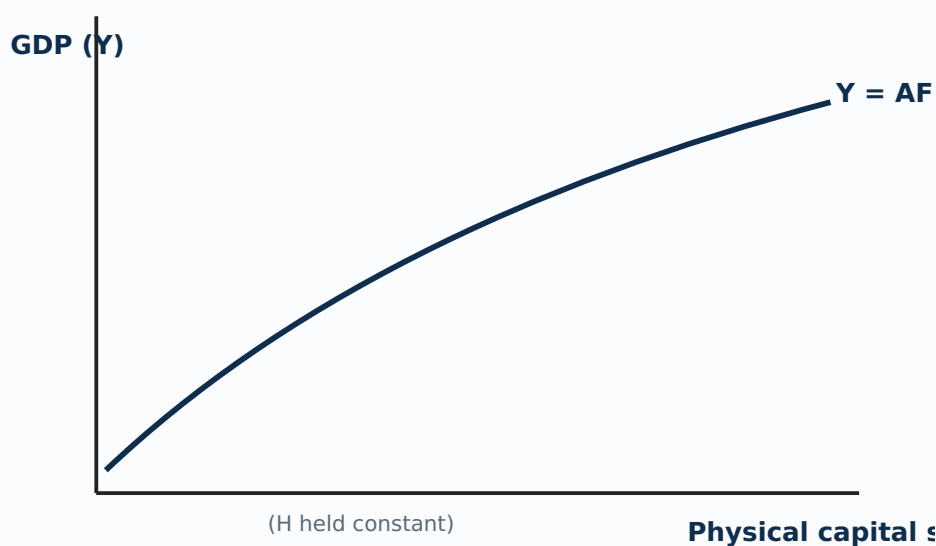


Figure 1: The aggregate production function with physical capital (K) on the horizontal axis and efficiency units of labor (H) held constant. The curve rises throughout — more capital always means more GDP — but becomes flatter as K increases, reflecting diminishing marginal product of capital.

Property 1: "More Is Better"

Increasing the physical capital stock, holding labor constant, always raises GDP. More machinery, equipment, and business structures give the existing (fixed) workforce more tools and infrastructure to work with, directly increasing what that workforce can produce. This is why the curve in Figure 1 rises throughout its length rather than eventually turning downward.

Property 2: The Law of Diminishing Marginal Product

Each additional unit of capital adds progressively less to GDP than the last. With efficiency units of labor fixed, each additional unit of capital has to be operated, on average, by the same fixed workforce — there's a limit to how much extra machinery a given number of workers can productively use at once. A factory that already has one machine per worker gains relatively little from adding a second identical machine per worker, compared to the gain from installing the very first one. This is why the curve becomes progressively flatter (concave) as K increases — its slope at any point is the marginal product of capital, and that marginal product steadily declines.

Implications

This underpins the Solow growth model's prediction of diminishing returns to capital accumulation alone. Because the marginal product of capital declines as K rises, simply accumulating more capital (through a higher saving rate, for instance) cannot sustain permanent GDP growth on its own — each successive round of investment adds less to output than the last, and the economy eventually approaches a steady state where additional capital investment merely offsets depreciation. Sustained long-run growth in GDP per capita instead requires ongoing technological progress (an increase in A) or growth in efficiency units of labor (H), not capital accumulation alone.

A higher capital stock still means a country is unambiguously more productive, all else equal. Cross-country differences in physical capital stock per worker are one of the three recognised sources of productivity differences across countries (alongside human capital and technology) — a country with a larger, well-maintained capital stock will, holding labor and technology constant, produce more GDP than an otherwise identical country with less capital, even though the marginal benefit of adding still more capital diminishes at high levels.

How the 100% Is Likely Allocated

Tier	Marks	What separates this tier
80-100%	80-100%	Correctly draws/describes an upward-sloping, concave curve, explains both defining properties ("more is better" and diminishing marginal product) with the underlying mechanism for each, and discusses the implication for sustained growth (capital accumulation alone cannot sustain it; technology/labor growth is needed).
50-79%	50-79%	Correctly describes the concave, upward-sloping shape and diminishing marginal product, but without the fuller discussion of growth implications.
20-49%	20-49%	Identifies GDP rises with capital but doesn't correctly capture the diminishing (concave) shape.
0-19%	0-19%	No meaningful attempt, or an incorrectly shaped or labelled diagram.

Question A2**100%**

Nobel laureate Simon Kuznets, who completed significant work on national income accounts in the 1930s, said that the welfare of a nation can scarcely be inferred from a measurement of national income.

Would you agree with this statement? Why or why not? Explain your answer.

FULL MODEL ANSWER

Agree — and there is a particular irony worth noting in the source of the quote: Simon Kuznets, the economist most responsible for developing the modern national income accounting framework that produces GDP, was himself explicit from the very beginning that his own creation was not a welfare measure. This isn't an external critique of GDP; it's a warning from its own architect.

Agree**Why GDP Cannot Be Read Directly as Welfare**

GDP omits large categories of activity and value entirely. It excludes depreciation of the physical capital stock and natural resources, home production (cooking, cleaning, childcare performed within a household rather than purchased on the market), transactions in the underground or informal economy, and leisure time — despite leisure being something people clearly value and would, in many circumstances, trade income for. A country where citizens work excessive hours with minimal leisure can register a higher GDP than an otherwise identical country with shorter working hours and more leisure, even though the second may plausibly represent higher welfare.

GDP does not net off negative externalities. Pollution, noise, and crime are not subtracted from national income — indeed, the clean-up after an environmental disaster or the private security spending driven by high crime rates adds to GDP (as production of clean-up or security services), even though the underlying events clearly reduce, not increase, genuine welfare.

GDP is silent on distribution. A rising GDP is entirely consistent with rapidly rising inequality, where the gains accrue overwhelmingly to a narrow segment of the population while typical living standards stagnate or fall. The aggregate figure carries no information about how broadly its growth is shared, yet how broadly growth is shared plausibly matters enormously for a nation's overall welfare.

GDP doesn't directly measure subjective wellbeing, health, or security.

The Easterlin Paradox — the finding that reported happiness does not rise in step with a country's rising average income once basic needs are met — is direct empirical evidence that income and welfare/happiness diverge in exactly the way Kuznets's statement warns against. GDP, as a pure measure of production, was never designed to capture happiness, health outcomes, personal security, or other dimensions widely understood to matter for a good life.

The Case for Some Qualification

GDP remains a genuinely useful, objective proxy, even if an imperfect one. GDP correlates strongly with many things people plausibly do value — access to healthcare, nutrition, education, and infrastructure — and it is calculated using a standardised, auditable methodology that permits consistent comparison across countries and over time, something few genuine welfare alternatives can match for regularity and comparability. Kuznets's own point was not that national income is worthless, but specifically that welfare “can scarcely be inferred” from it — a caution about over-interpreting the measure, not a claim that it has no informational value at all.

A strong answer explicitly engages with the irony that Kuznets himself — the person who built the modern national accounting framework — issued this warning, since it lends particular authority to the statement, and distinguishes between GDP being a poor welfare measure versus GDP still being a useful production/activity measure for many policy purposes.

How the 100% Is Likely Allocated

Tier	Marks	What separates this tier
80-100%	80-100%	Takes a clear position (agreeing, with reasoned qualification), supports it with multiple specific mechanisms (omitted categories, externalities, distribution, well-being/Easterlin Paradox), and shows critical balance by acknowledging GDP's genuine strengths as an objective, comparable production measure.
50-79%	50-79%	Takes a position and supports it with two or three specific mechanisms, but without the fuller balanced discussion of GDP's genuine strengths.
20-49%	20-49%	General, largely unsupported agreement or disagreement with limited specific reasoning.
0-19%	0-19%	No meaningful attempt, or a fundamental misunderstanding of what GDP measures.

Question A3

100%

The following table lists 2012 GDP per capita for four countries. The data are given in the national currencies of the countries. It also lists the price of a Big Mac burger in local currency in each country in 2012.

	2012 GDP per Capita	2012 Big Mac Price
Norway (krone)	579,162	41 krone
Poland (zloty)	41,398	9.1 zloty
Turkey (Turkish lira)	19,580	6.6 Turkish lira
United Kingdom (British pound)	24,740	2.49 pounds

The price of a Big Mac in the United States in 2012 was \$4.55.

- i. Using the Big Mac burger as a representative commodity common to the countries, calculate the purchasing power parity (PPP)-adjustment factor for each country, and then the PPP level of per capita GDP in each country.
- ii. What are the disadvantages of using Big Macs to measure purchasing power parity?

FULL MODEL ANSWER

Part (i) — PPP-Adjustment Factors and PPP GDP per Capita

The Big Mac index uses a single, broadly standardised product sold in many countries as a stand-in for a full basket of goods. The PPP-adjustment factor for each country is the ratio of the local Big Mac price to the U.S. Big Mac price — this represents the exchange rate at which a Big Mac would cost the same in both countries, i.e. the implied purchasing-power-parity exchange rate.

PPP-adjustment factor = Local Big Mac price ÷ US Big Mac price (\$4.55)

PPP GDP per capita = Local GDP per capita ÷ PPP-adjustment factor

Country	PPP-Adjustment Factor	PPP GDP per Capita
Norway	$41 \div 4.55 = 9.011 \text{ krone}/\$$	$579,162 \div 9.011 \approx \$64,273$
Poland	$9.1 \div 4.55 = 2.000 \text{ zloty}/\$$	$41,398 \div 2.000 \approx \$20,699$
Turkey	$6.6 \div 4.55 = 1.4505 \text{ lira}/\$$	$19,580 \div 1.4505 \approx \$13,498$
United Kingdom	$2.49 \div 4.55 = 0.5473 \text{ £}/\$$	$24,740 \div 0.5473 \approx \$45,208$

It's worth sense-checking these implied rates against actual market exchange rates. The average USD/NOK market exchange rate in 2012 was around 5.82 krone per dollar — well below the Big Mac-implied rate of 9.011 krone per dollar calculated above. This gap means a Big Mac cost considerably more in Norway than the market exchange rate alone would predict, consistent with Norway's well-known reputation for a high cost of living, and illustrates exactly the local-cost distortion problem discussed in part (ii): a large share of the Big Mac's price in Norway reflects high local rents and wages rather than the cost of internationally tradable inputs.

Part (ii) — Disadvantages of Using Big Macs to Measure PPP

A Big Mac is not a tradable good. Classical purchasing power parity theory relies on arbitrage in internationally tradable goods driving prices toward equality once converted to a common currency. A Big Mac cannot be bought in one country and resold in another, so there is no arbitrage mechanism forcing its price toward the parity level — its price can and does diverge from what a genuine tradable-goods PPP measure would show.

Big Mac prices are heavily influenced by local non-tradable costs. A large share of a Big Mac's price reflects local rent, labor costs, and local taxes (VAT/sales tax) rather than the cost of the tradable ingredients themselves — countries with high local rents or wages will show an inflated Big Mac price even if their broader price level for genuinely tradable goods is similar to other countries, biasing the PPP-adjustment factor.

A single product is not representative of a typical consumption basket. PPP is properly meant to reflect the cost of a broad, representative basket of goods and services that people actually consume. A Big Mac is one narrow product from one specific industry (fast food), and may not be representative at all in countries where beef is not commonly eaten, where fast food is considered a luxury or niche good rather than an everyday purchase, or where dietary and cultural patterns differ substantially from the U.S. market the product was designed around.

McDonald's may price strategically rather than reflecting local costs directly. Pricing can reflect deliberate market positioning or price discrimination by McDonald's in different countries (charging what the local market will bear, rather than passing through local costs one-for-one), and franchise structures, import tariffs on specific ingredients, and local competitive conditions can all distort the price in ways unrelated to the general local price level PPP is meant to capture.

How the 100% Is Likely Allocated

Tier	Marks	What separates this tier
80-100%	80-100%	All four PPP-adjustment factors and PPP GDP per capita figures calculated correctly with clear working shown, AND at least three genuine disadvantages of the Big Mac method explained with mechanism (non-tradability, local cost composition, non-representativeness, strategic pricing).
50-79%	50-79%	Most calculations correct with minor arithmetic slips, and two or three disadvantages identified with reasonable explanation.
20-49%	20-49%	Significant calculation errors, or disadvantages named without explanation of the underlying mechanism.
0-19%	0-19%	No meaningful attempt, or a fundamentally incorrect method (e.g. dividing the wrong way round).

Section B — Long Questions

Answer 1 of 2 · 60% of total marks · parts unevenly weighted as shown

Question B1

Growth History, Institutions & Entrepreneurship

B1 (a)

30%

The graph below shows an index of world GDP per capita from 1000 B.C. to the year 2000.

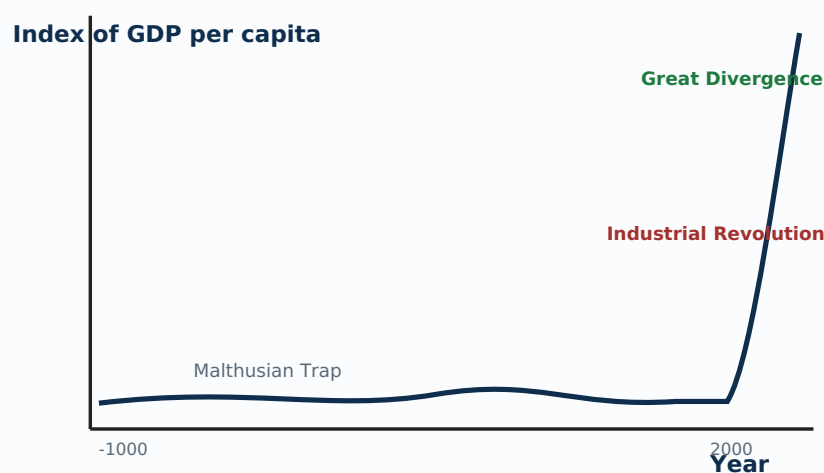


Figure 1: World GDP per capita, 1000 B.C. to 2000. Income stays close to a flat subsistence level for millennia (the Malthusian Trap), before rising sharply from around the mid-eighteenth century (the Industrial Revolution) and accelerating further into the Great Divergence.

As you can see, over most of that period, global economic growth was virtually nonexistent. While there were periods that experienced some increase in per capita income, sustained growth begins only in the mid-eighteenth century, and explodes after that — by the year 2000, income per capita is 12 times what it had been 250 years before.

Required: Explain what accounts for such a dramatic change in economic growth beginning in the eighteenth century.

FULL MODEL ANSWER

For nearly three thousand years, the world economy was caught in what economic historians call the **Malthusian Trap**: any temporary rise in income per capita above subsistence level was reliably reversed by higher fertility and population growth, driving income back down. Escaping this trap required two specific, mutually reinforcing developments that began in Britain in the mid-eighteenth century.

1. The Malthusian Trap, and Why It Held for So Long

Thomas Malthus's mechanism. Writing in 1798, Malthus argued that any rise in income per capita above the subsistence level would lead to higher fertility rates; the resulting population growth would then drive per capita income back down to subsistence, since a larger population had to share essentially the same, technologically static amount of land and output. Without ongoing technological advance to break this cycle, pre-modern economies were structurally locked into little to no sustained growth in income per capita, exactly as Figure 1 shows for the roughly 2,800 years before 1800.

2. What Finally Broke the Cycle: Two Reinforcing Trends

First, the Industrial Revolution created genuine, sustained technological progress. The introduction of new machines and methods of production — beginning in textile manufacturing and then spreading into other production processes and transportation — created technological progress on a scale and pace never seen before. Crucially, this progress didn't just produce a one-off jump in output; it proved to be self-sustaining, with each new innovation (steam power, mechanised production, later the railway) enabling further innovation, allowing output to grow faster than population for the first time in a genuinely durable way.

Second, the demographic transition broke the Malthusian fertility

response. As people moved from rural agriculture into urban manufacturing, fertility rates fell rather than rising as the older Malthusian pattern would predict — children shifted from being economically productive farm labour to a costly, education-intensive investment in an urban, industrialising economy, reducing the incentive for large families. This demographic transition is precisely what allowed rising income to actually *stick* this time, rather than being reabsorbed by population growth as it always had been before.

Why Both Trends Were Necessary Together**Technology alone would not have been sufficient without the fertility**

response. Had industrial technology advanced without any corresponding change in fertility behaviour, the classic Malthusian mechanism would likely have simply reasserted itself — a larger population absorbing the productivity gains and driving per capita income back toward subsistence, exactly as had happened after every previous episode of technological improvement (see the smaller, temporary bumps visible in Figure 1 before 1800). It's the combination of sustained technological progress *and* falling fertility that explains why growth from the eighteenth century onward was different in kind, not just in degree, from anything that came before — producing the sharp, sustained “hockey stick” pattern and the subsequent Great Divergence in income levels across countries visible in the data.

How the 30% Is Likely Allocated

Tier	Marks	What separates this tier
24-30%	24-30%	Correctly explains the Malthusian Trap mechanism for the pre-1800 stagnation, identifies BOTH the Industrial Revolution (technological progress) and the demographic transition (falling fertility) as the two trends responsible for breaking it, and explains why both were necessary together rather than either alone being sufficient.
15-23%	15-23%	Correctly identifies the Industrial Revolution and/or demographic transition but without the fuller explanation of the Malthusian mechanism they were breaking, or without explaining why both were needed together.
6-14%	6-14%	Vague reference to "technology improved" without identifying the demographic transition or explaining the underlying Malthusian mechanism.
0-5%	0-5%	No meaningful attempt.

B1 (b)**30%**

Look at the following map of Nogales, a twin city that is divided by the U.S. border. One part of Nogales lies in the United States, in Arizona, and the other part lies in Sonora, Mexico. Life in Nogales, Mexico is very different from life in Nogales, Arizona. The average income in Nogales, Mexico is about one-third the average income in Nogales, Arizona. Education levels, life expectancy, and health conditions are better in Nogales, Arizona than in Nogales, Mexico. Unlike Nogales in Arizona, Nogales in Mexico has only recently adopted political reforms, bringing it closer to functioning as a democracy. Crime rates are also lower in Nogales, Arizona than in Nogales, Mexico. Since both cities are located so close to each other, they share similar geographical conditions and climate. The inhabitants of both cities also share a common ancestry and enjoy the same types of food and music.

Required: Based on this information and your own research, what factors do you think can explain why Nogales, Arizona is so much more prosperous than Nogales, Mexico?

FULL MODEL ANSWER

The Nogales case is a textbook natural experiment: geography, climate, ancestry, culture, food, and music are all held constant across the border — the two cities sit within sight of each other. What differs is entirely on the other side of the ledger, and that is precisely the point the example is designed to make. The explanation lies almost entirely in **institutions**, not in geography or culture.

Defining Institutions

Douglass North's definition. Economic historian Douglass North defined institutions as “the rules of the game in a society, or, more formally, the humanly devised constraints that shape human interaction.” Institutions include a country's legal system, property rights protections, electoral and political systems, regulatory bodies, and the informal norms (trust, corruption tolerance, rule-following) that govern how people and firms actually behave.

Why Institutions, Not Geography or Culture, Explain the Gap

The geography/culture explanations are directly ruled out by the setup of the question itself. Nogales, Arizona and Nogales, Mexico share the same climate, terrain, natural resources, ancestry, food, and music — the very factors that geographic or cultural theories of development would point to. Since these are identical on both sides of the border, they cannot be what's driving a threefold income gap between two halves of literally the same city. This is exactly the kind of comparison Daron Acemoglu and James Robinson use in *Why Nations Fail* (2012) to argue that institutions, not geography or culture, are the fundamental cause of prosperity differences — the Nogales case is one of their own central illustrations, alongside the Korean peninsula (see below).

Property rights and the rule of law. In Nogales, Arizona, the U.S. legal system provides strong, reliably enforced property rights and contract enforcement — an entrepreneur can invest, build a business, and trust that courts will protect their ownership and enforce agreements. Where property rights are weak or inconsistently enforced, as historically in much of Mexico, the incentive to invest, start a business, or accumulate capital is undermined, since the returns to that investment are less secure.

Political institutions and democratic accountability. The passage notes Nogales, Mexico “has only recently adopted political reforms, bringing it closer to functioning as a democracy” — a more recently and less deeply embedded democratic system typically comes with weaker checks on the executive, a shorter track record of peaceful transitions of power, and institutions still in the process of building credibility. Acemoglu and Robinson's framework distinguishes **inclusive** institutions (which broadly distribute political power and economic opportunity, incentivising investment and innovation) from **extractive** institutions (which concentrate power and resources among a narrow elite) — the U.S. side has a much longer history of inclusive political institutions than the Mexican side.

Crime, corruption, and the rule of law's enforcement in practice. Lower crime rates in Nogales, Arizona reflect (and reinforce) stronger law enforcement institutions and judicial capacity. High crime and, historically, drug-cartel-related violence on the Mexican side of the border directly deter investment, raise the cost of doing business (security costs, insurance, disrupted supply chains), and discourage entrepreneurship — a direct institutional channel through which weak rule of law translates into lower prosperity.

Access to U.S. markets, capital, and regulatory institutions. Firms and residents in Nogales, Arizona automatically operate within the U.S. federal legal, financial, and regulatory system — deep and liquid capital markets, federally insured banking, and federal infrastructure investment — none of which residents on the Mexican side have equal access to, despite being only metres away. Even identical entrepreneurial talent and effort produce very different outcomes depending on which side of the institutional boundary it's deployed on.

A Parallel Case Strengthens the Argument

North and South Korea. The clearest parallel supporting the institutions hypothesis is the Korean peninsula: a single people, language, culture, and (until 1948) shared history and geography, split into two dramatically different economic outcomes by nothing other than the institutions each half subsequently adopted — South Korea's market-oriented, increasingly inclusive institutions versus North Korea's centrally planned, extractive regime. As with Nogales, every plausible geographic or cultural explanation is held constant, leaving institutional divergence as the explanation that actually varies alongside the outcome.

A strong answer explicitly uses the logic of the comparison (what's held constant vs. what differs) rather than just listing institutional factors in the abstract — the power of the Nogales example is precisely that it isolates institutions as the variable that changes while geography and culture do not.

How the 30% Is Likely Allocated

Tier	Marks	What separates this tier
24-30%	24-30%	Correctly identifies institutions (property rights, political institutions/democracy, rule of law, crime/corruption, access to capital and regulatory systems) as the explanation, explicitly uses the comparison logic (geography/culture held constant, institutions vary), cites the Acemoglu/Robinson inclusive-vs-extractive framework, and draws the North/South Korea parallel or an equivalent supporting case.
15-23%	15-23%	Correctly identifies institutions as the key driver with several specific mechanisms, but without explicitly using the comparison logic or citing the relevant literature.
6-14%	6-14%	Vague reference to "government" or "policy" differences without specific institutional mechanisms, or an answer that leans on geography/culture despite the question's setup ruling these out.
0-5%	0-5%	No meaningful attempt, or an answer entirely based on factors the question explicitly states are held constant between the two cities.

B1 (c)**40%**

Suppose the return to entrepreneurship and the opportunity cost of entrepreneurship curves are described by the following equations:

$$R = 500,000 - 100N$$

$$C = 100,000 + 300N$$

Where: R = returns to entrepreneurship, C = cost of entrepreneurship, N = number of entrepreneurs

- i. Based on the equations given, plot the curves and explain their shapes. Identify the equilibrium number of entrepreneurs in this economy, and the equilibrium returns to entrepreneurship on the graph.

FULL MODEL ANSWER**Shape of the Two Curves**

R = 500,000 – 100N is downward-sloping. The return to entrepreneurship falls as more people become entrepreneurs, because the most promising business opportunities and market niches tend to be taken up first — each additional entrepreneur faces more competition for customers, funding, and talent than the entrepreneurs who came before them, so the marginal return to entering entrepreneurship declines as N rises.

C = 100,000 + 300N is upward-sloping. The opportunity cost of entrepreneurship — what a potential entrepreneur gives up by not pursuing their next-best alternative (typically stable employment) — rises as more people become entrepreneurs, because as increasingly higher-skilled or higher-earning individuals are drawn into entrepreneurship, the value of what they're giving up (their next-best outside option) is progressively higher.

Finding the Equilibrium

Equilibrium occurs where $R = C$:

$$500,000 - 100N = 100,000 + 300N$$

$$400,000 = 400N$$

$N^* = 1,000$ entrepreneurs

$$R^* = 500,000 - 100(1,000) = 400,000$$

$$\text{Check: } C^* = 100,000 + 300(1,000) = 400,000 \checkmark$$

Equilibrium return/cost = €400,000

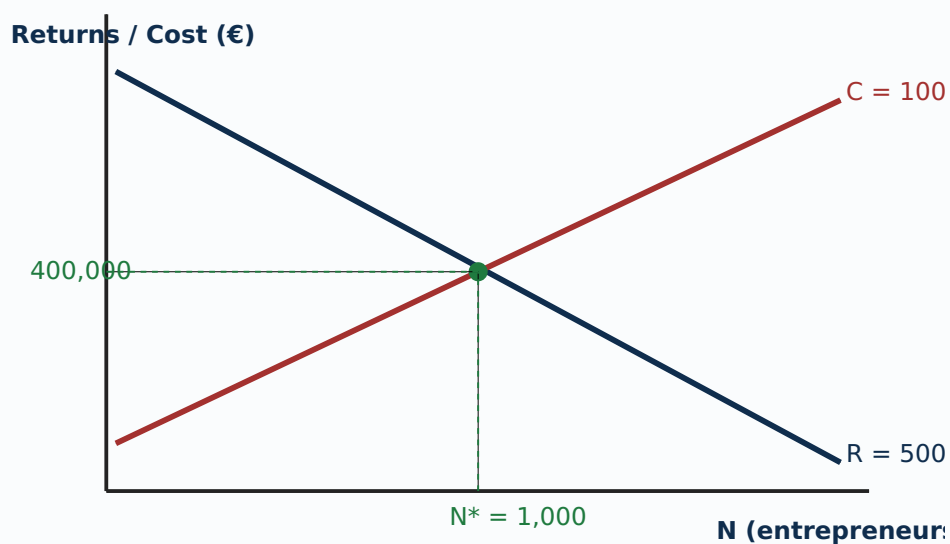


Figure 2: The return to entrepreneurship (R) falls, and the opportunity cost of entrepreneurship (C) rises, as the number of entrepreneurs (N) increases. Equilibrium occurs where the two curves intersect: $N^* = 1,000$, at a return/cost of €400,000.

Why this point is the equilibrium. At any N below 1,000, the return to entrepreneurship exceeds its opportunity cost ($R > C$), so additional people are drawn into entrepreneurship, pushing N up. At any N above 1,000, the opportunity cost exceeds the return ($C > R$), so entrepreneurs exit toward their next-best alternative, pushing N back down. $N^* = 1,000$ is the only point where neither entry nor exit is individually rational, making it the stable equilibrium.

How This Part of the 40% Is Likely Allocated (roughly half)

Tier	Marks	What separates this tier
16-20%	16-20%	Correctly explains why R slopes down and C slopes up (competition for opportunities; rising opportunity cost of higher-value alternative workers), correctly solves for $N^* = 1,000$ and the equilibrium value of €400,000 with full working shown, and correctly plots/labels both on the graph with the equilibrium point identified.
10-15%	10-15%	Correct equilibrium calculation but with limited explanation of why each curve has its particular slope.
4-9%	4-9%	Attempts the algebra but with an error in solving for N^* , or no graph.
0-3%	0-3%	No meaningful attempt.

B1 (c)(ii)

ii. Due to an ageing population, the government encourages more people to work by lowering the retirement age and providing more job-matching and retraining services. What is the effect of such policies on the equilibrium number of entrepreneurs? Use the graph in (a) to answer the question.

FULL MODEL ANSWER

The question describes the government as encouraging more people to work “by lowering the retirement age” — but lowering the retirement age would let people retire earlier, reducing labour force participation, the opposite of the stated goal. This is treated as a likely typo for “raising the retirement age” throughout this answer, since that's the only reading consistent with encouraging more people to work amid an ageing population.

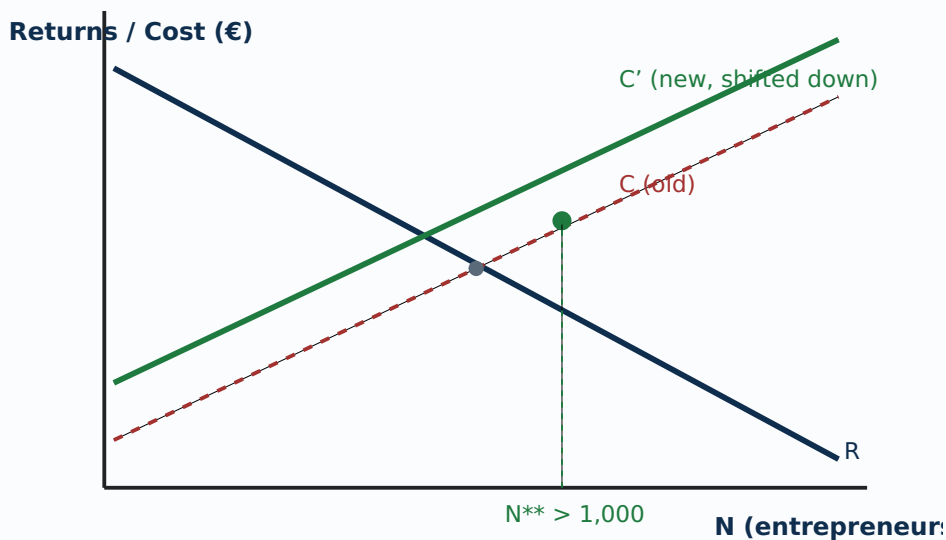


Figure 3: Job-matching and retraining services reduce the opportunity cost of entrepreneurship at every level of N , shifting the C curve down/right from C to C' . The new equilibrium sits further right, with a higher number of entrepreneurs (N^{**}) and a lower equilibrium return.

Job-matching and retraining services reduce the opportunity cost of entrepreneurship.

Better job-matching lowers the search costs and uncertainty involved in leaving stable employment to pursue entrepreneurship, and retraining services lower the cost of transitioning skills into a new entrepreneurial venture. Both effects reduce the opportunity cost, C , associated with becoming an entrepreneur at every level of N — shifting the entire C curve downward (equivalently, rightward).

A raised retirement age adds more potential entrepreneurs to the pool.

Keeping older workers economically active for longer (rather than retiring) increases the overall pool of people available to become entrepreneurs, reinforcing the same directional effect on the equilibrium.

Effect on equilibrium: the number of entrepreneurs rises, and equilibrium returns fall.

With the C curve shifted down while R is unchanged, the new intersection point sits further to the right along the (unchanged) R curve — the equilibrium number of entrepreneurs increases beyond $N^* = 1,000$, while the equilibrium return (read off the R curve, which is downward-sloping) falls below the original €400,000, since supporting more entrepreneurs means moving further down the R curve.

How This Part of the 40% Is Likely Allocated (roughly a quarter)

Tier	Marks	What separates this tier
8-10%	8-10%	Correctly identifies that job-matching/retraining shifts the C curve down (not the R curve), correctly identifies the direction of the resulting change in both N and equilibrium returns, and illustrates this using the graph from part (i).
5-7%	5-7%	Correctly identifies more entrepreneurs result, but without correctly identifying which curve shifts or explaining the effect on equilibrium returns.
2-4%	2-4%	Vague reference to "more entrepreneurs" without using the C/R framework at all.
0-1%	0-1%	No meaningful attempt.

B1 (c)(iii)

iii. Continuing from part (ii), the government seeks to further improve the legal environment. What would the equilibrium number of entrepreneurs be? What would the effect on GDP be?

FULL MODEL ANSWER

A better legal environment raises the return to entrepreneurship at every level of N . Stronger property rights and more reliable contract enforcement reduce the risk that an entrepreneur's investment, ideas, or contractual agreements will be expropriated, stolen, or unenforceable — raising the expected, risk-adjusted return to starting a business at every level of N . This shifts the R curve **upward** (Figure 4), on top of the C curve already shifted down from part (ii).

The new equilibrium number of entrepreneurs. With R shifted up and C already shifted down, the new intersection point moves further to the right than either shift would produce alone — the equilibrium number of entrepreneurs, N^{***} , is higher again than N^{**} from part (ii), which was itself already above the original $N^* = 1,000$. The precise numerical value depends on the specific magnitude of the legal-environment improvement, which isn't quantified in the question, so the answer should be given directionally: an even higher equilibrium number of entrepreneurs than in part (ii).

The effect on GDP: unambiguously positive. More entrepreneurs, operating in an environment where their higher returns reflect genuinely more secure and productive investment (rather than merely more entrepreneurs competing over the same fixed opportunities), translates into more business formation, more innovation, and more efficient allocation of resources toward higher-return activities — all of which raise aggregate output. This connects directly to the institutions-and-growth theme running through this paper (see part (b)): a stronger legal environment is exactly the kind of institutional improvement — alongside secure property rights and reliable contract enforcement — that raises a country's GDP by raising the productive return to economic activity.

A strong answer explicitly distinguishes the mechanism in part (iii) (raising R, via reduced risk/increased security of the returns to entrepreneurship) from the mechanism in part (ii) (lowering C, via reduced search/transition friction) — both increase the equilibrium number of entrepreneurs, but through genuinely different channels, and a complete answer shows awareness of both moving in this final part rather than only discussing one.

How This Part of the 40% Is Likely Allocated (roughly a quarter)

Tier	Marks	What separates this tier
8-10%	8-10%	Correctly identifies that a better legal environment shifts R upward (distinct from the C-curve shift in part (ii)), correctly identifies the new equilibrium has even more entrepreneurs than part (ii), and correctly reasons that GDP rises as a result, tying back to the institutions-and-growth theme.
5-7%	5-7%	Correctly identifies more entrepreneurs and higher GDP but without clearly distinguishing the R-curve shift from the C-curve shift in part (ii).
2-4%	2-4%	Vague reference to "better legal environment helps growth" without using the R/C framework.
0-1%	0-1%	No meaningful attempt.

Question B1 Total: 100% (60% of the Paper)

Question B2

Business Cycles, Multipliers & Stabilisation Policy

B2 (a)

20%

(a) What are the major sources of fluctuations identified by each of these three schools of thought (i) real business cycle theory (ii) Keynesian theory, and (iii) financial and monetary theories?

FULL MODEL ANSWER

Macroeconomists have identified three broad schools of thought explaining what actually drives short-run economic fluctuations, each emphasising a different underlying source of shocks.

(i) Real Business Cycle Theory

Source of fluctuations: productivity and technology shocks. Real business cycle theory attributes fluctuations to changes in productivity and technology. A wave of productivity-enhancing innovation causes an economic expansion, as firms can produce more output from the same inputs. Conversely, a rise in the price of a key input — oil being the classic example — acts as a negative supply shock and can trigger a recession, since it raises production costs economy-wide independent of demand conditions.

(ii) Keynesian Theory

Source of fluctuations: shifting expectations and "animal spirits." Keynesian theory focuses on changes in business and consumer expectations about the future. "Animal spirits" — John Maynard Keynes's term for the psychological factors driving shifts in business and consumer mood or sentiment — can independently trigger either decreases in spending (recessions) or increases in spending (expansions), regardless of whether any change in underlying productivity or fundamentals has actually occurred. A negative shock can generate pessimism; willingness to spend falls, and — critically for the Keynesian mechanism — that initial fall in spending is not offset by increased spending elsewhere in the economy, and is instead amplified further through multiplier effects as one person's reduced spending becomes another person's reduced income.

(iii) Financial and Monetary Theory

Source of fluctuations: changes in the money supply, prices, and interest rates. Associated principally with Milton Friedman, this view attributes fluctuations to monetary factors: a decrease in the money supply (M2) causes the price level to fall, which reduces employment because wages are downwardly rigid (they don't easily adjust down to match falling prices, so real labour costs effectively rise, and firms cut jobs). A falling money supply also raises the real interest rate, which reduces investment spending by firms, compounding the contractionary effect.

A strong answer keeps the three theories genuinely distinct rather than blending them — RBC theory locates the shock in the real, productive side of the economy (technology/productivity); Keynesian theory locates it in psychology and expectations (demand-side, amplified by multipliers); and financial/monetary theory locates it in the money supply and interest rates. Real-world recessions are often explained using elements of more than one theory simultaneously, which is a reasonable point to make, but the theories themselves are analytically separate.

How the 20% Is Likely Allocated

Tier	Marks	What separates this tier
16-20%	16-20%	Correctly identifies the distinct source of fluctuations for all three schools (RBC: productivity/technology shocks; Keynesian: animal spirits/expectations amplified by multipliers; financial/monetary: money supply, price level, and interest rates), with the specific mechanism explained for each rather than just named.
10-15%	10-15%	Correctly identifies two of the three theories with reasonable mechanism detail, the third vague or missing.
4-9%	4-9%	Names the three schools but without correctly distinguishing their underlying mechanisms.
0-3%	0-3%	No meaningful attempt.

B2 (b)**20%**

(b) The concept of multipliers was one of the key elements of John Maynard Keynes's theory of fluctuations. What is a multiplier? Explain with an example.

FULL MODEL ANSWER

A **multiplier** is the mechanism by which an initial change in spending generates a larger total change in economy-wide income and output, because one person's spending becomes another person's income, which is then partly re-spent, generating further rounds of income and spending down the chain.

The Mechanism

Spending circulates and compounds, rather than stopping after one transaction. When one euro of new spending occurs, the recipient of that euro (a worker, a supplier, a landlord) doesn't save it entirely — they spend a portion of it themselves, which becomes income for someone else, who in turn spends a portion of that, and so on. The total increase in economy-wide income ends up larger than the initial injection of spending, with the exact size of the multiplier depending on how much of each round is re-spent versus saved or spent on imports (which leak out of the domestic spending chain).

Multipliers work in both directions, and are central to Keynesian theory precisely because they amplify shocks. An initial fall in spending (driven, say, by a negative shift in animal spirits) doesn't just directly reduce the income of whoever was going to receive that spending — it reduces their spending too, which reduces someone else's income, and so on, amplifying the original shock into a much larger overall contraction. This amplification is exactly why Keynesian theory treats multipliers as one of the central elements explaining how economic fluctuations can become so much larger than the shocks that triggered them.

A Worked Example: The Páirc Uí Chaoimh Redevelopment

The initial injection. Consider the redevelopment of Páirc Uí Chaoimh, the GAA stadium in Cork. The construction spend itself is the initial injection — wages paid to construction workers, payments to building material suppliers, and fees paid to architects and contractors.

The first round of re-spending. The construction workers who receive those wages don't save all of it — they spend a good share of it locally: in Cork city cafes, shops, and pubs near the site, on transport, and on other everyday goods and services. That spending becomes income for the owners and staff of those local businesses.

Further rounds, and the eventual total effect. Those café and shop owners, in turn, spend part of their increased income on their own suppliers, staff wages, and purchases — and so the chain continues, each round smaller than the last as some income is saved or spent outside the local area (a “leakage”) rather than re-spent locally. The eventual total increase in economic activity generated across Cork city, once every round of this chain is added up, is larger than the construction spend alone — this total effect (original spend plus all the induced rounds of re-spending) is the multiplier effect, and is exactly the kind of local economic impact argument regularly used to justify major stadium and infrastructure redevelopments.

A complete answer explains *why* the chain eventually stops (leakages to saving, taxation, and imports mean each round is progressively smaller, so the total effect converges to a finite multiple of the initial spend rather than growing without limit) rather than simply asserting spending “multiplies” without explaining the mechanism or its limit.

How the 20% Is Likely Allocated

Tier	Marks	What separates this tier
16-20%	16-20%	Correctly defines the multiplier mechanism (spending becomes income, partly re-spent, generating further rounds), explains why the effect eventually diminishes and converges (leakages to saving/imports), and applies it in a fully worked, specific example (such as Páirc Uí Chaoimh) tracing at least two rounds of the chain.
10-15%	10-15%	Correctly defines the multiplier concept with a relevant example, but without tracing multiple rounds or explaining why the effect eventually diminishes.
4-9%	4-9%	Vague reference to spending having “knock-on effects” without a clear mechanism or worked example.
0-3%	0-3%	No meaningful attempt.

B2 (c)**30%**

(c) What do countercyclical policies mean? During an economic boom, what countercyclical fiscal and monetary policies are used? Please use a diagram of the business cycle to help explain your answer.

FULL MODEL ANSWER

Countercyclical policies are fiscal and monetary policy actions deliberately taken to move *against* the current direction of the business cycle — restraining the economy during a boom (when it is running above trend, as in Figure 2) and stimulating it during a recession (when it is running below trend) — in order to smooth out fluctuations and keep actual GDP closer to its trend path, rather than reinforcing whichever direction it happens to be moving in already.

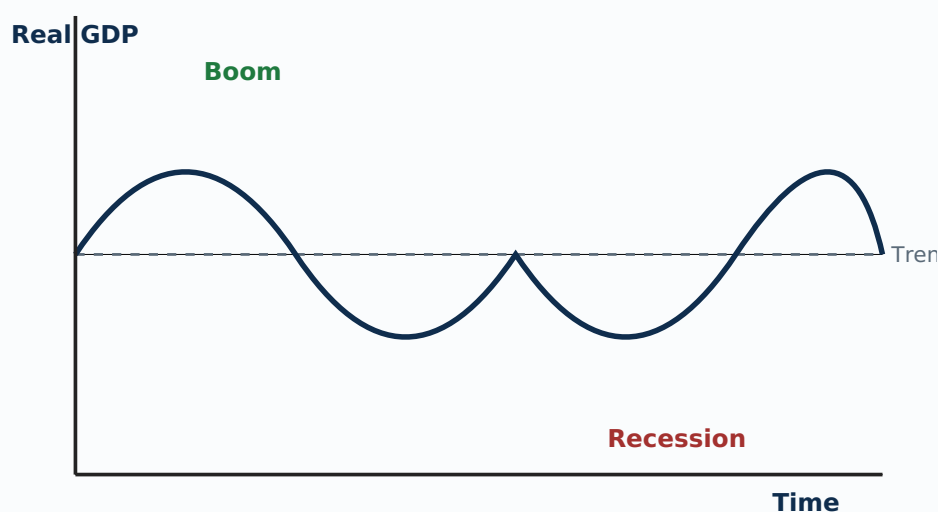


Figure 2: The business cycle — actual real GDP (solid line) fluctuating around its long-run trend (dashed line). A boom is a period where GDP rises above trend; a recession is a period where it falls below trend.

Countercyclical Policy During a Boom

The logic: cool down an economy that is overheating. During a boom, GDP is running above its sustainable trend, which risks fuelling inflation, asset bubbles, and unsustainable borrowing. Countercyclical policy during a boom aims to withdraw some stimulus from the economy specifically to prevent these excesses, rather than letting the boom run unchecked toward a sharper eventual correction.

Countercyclical Fiscal Policy in a Boom

Reduce government spending and/or raise taxes. A government pursuing countercyclical fiscal policy during a boom would restrain its own spending growth and/or raise taxes, withdrawing some private demand from an already-strong economy — the opposite of the spending increases/tax cuts used to fight a recession. In practice this also means resisting the temptation to spend windfall tax revenues (which rise automatically during a boom as incomes and profits rise) and instead saving them, for instance into a sovereign wealth or rainy-day fund, for use during the next downturn.

Countercyclical Monetary Policy in a Boom

Raise interest rates. A central bank pursuing countercyclical monetary policy during a boom raises interest rates, which increases the cost of borrowing for both households and firms, cooling consumption and investment spending, and helping contain inflationary pressure building up from the strong economy — the reverse of the interest rate cuts (and, in more extreme downturns, quantitative easing) used to stimulate a recession.

A common error is describing expansionary policies (spending increases, tax cuts, rate cuts) when asked specifically about boom-time countercyclical policy — those are the correct countercyclical response to a *recession*, precisely the opposite direction to what's being asked here. Countercyclical, by definition, means pushing against the current phase of the cycle: contractionary policy in a boom, expansionary policy in a recession.

How the 30% Is Likely Allocated

Tier	Marks	What separates this tier
24-30%	24-30%	Correctly defines countercyclical policy as moving against the current cycle phase, correctly identifies contractionary fiscal policy (spending restraint/tax increases) AND contractionary monetary policy (interest rate rises) as the boom-time response, and uses a correctly labelled business cycle diagram distinguishing boom from recession relative to trend.
15-23%	15-23%	Correctly identifies contractionary fiscal and monetary policy for a boom but without a clear diagram, or with one of the two policy types missing.
6-14%	6-14%	Describes expansionary (recession-fighting) policies instead of the boom-time contractionary policies actually asked for.
0-5%	0-5%	No meaningful attempt.

B2 (d)**30%**

(d) Critically evaluate the fiscal and monetary policies of the Irish government since the turn of the 21st century.

FULL MODEL ANSWER

Since Ireland adopted the euro in 1999, its own government has had no independent monetary policy lever at all — interest rates are set for the whole eurozone by the European Central Bank (ECB). This answer therefore evaluates Irish fiscal policy (which Ireland fully controls) across four distinct periods, and evaluates monetary policy specifically in terms of how well a single ECB rate has fitted Ireland's own economic conditions at each stage.

2000–2008: The Celtic Tiger Boom — Pro-Cyclical Fiscal Policy

Fiscal policy reinforced the boom rather than restraining it. Rather than pursuing countercyclical policy (as discussed in part (c)) by restraining spending and banking windfall revenue during the boom, Irish fiscal policy through the 2000s was broadly **pro-cyclical**: government spending and tax cuts both expanded significantly, funded substantially by transaction-based taxes (stamp duty, VAT on construction) tied directly to an overheating property market. This left the exchequer's revenue base structurally exposed to a property market correction.

Monetary policy: a single ECB rate that was a poor fit for an overheating Ireland. As part of the eurozone, Ireland faced the same ECB policy rate as much larger, more sluggish economies like Germany and France. Those rates were arguably too low for an Irish economy running well above its sustainable trend — a textbook “one size fits none” problem of monetary union, since the single rate could not simultaneously suit an overheating Ireland and a stagnating German economy.

2008–2013: Financial Crisis, Bank Guarantee, and the Bailout

A blanket bank guarantee and severe fiscal consolidation. Following the property crash and global financial crisis, the government issued a blanket guarantee of the banking sector's liabilities in September 2008, and subsequently pumped tens of billions of euros into recapitalising and nationalising banks (including Anglo Irish Bank), pushing the sovereign's own debt burden to unsustainable levels. Ireland formally entered an €85 billion EU-IMF-ECB ("Troika") assistance programme in December 2010, exiting in December 2013 having implemented severe fiscal consolidation (spending cuts and tax rises, including a 14% cut to public sector pay in 2009) as a condition of the programme.

Evaluation: necessary, but arguably compounded a downturn that pro-cyclical boom-time policy had made worse. The scale of austerity required was directly a consequence of the fiscal position built up (and left exposed) during the pro-cyclical boom years — a clear illustration of why countercyclical fiscal discipline during a boom matters, since its absence left far less room to manoeuvre once the downturn hit. On the monetary side, Ireland had no independent lever to cushion the shock at all (no ability to devalue its own currency or set its own interest rate), leaving fiscal austerity as essentially the only adjustment tool available.

2014-2019: Recovery

A strong recovery, aided in part by ECB accommodation. Ireland's economy recovered strongly through this period, with unemployment falling from a peak of over 15% in 2011-2012 to under 6% by 2019. The ECB's own quantitative easing programme (asset purchases) from 2015 kept borrowing costs low across the eurozone, indirectly supporting Ireland's recovery and debt sustainability even though Ireland had no direct control over that policy.

2020-Present: COVID-19, and a Return of Fiscal Windfalls

A large, temporary fiscal expansion during COVID-19. The government ran significant deficits to fund the Pandemic Unemployment Payment, the Employment Wage Subsidy Scheme, and broader health and business supports — a genuinely countercyclical fiscal response to an exceptional shock, in contrast to the pro-cyclical pattern of the 2000s.

Since 2022-2023: a corporate tax windfall, and the return of an old lesson only partially learned. Ireland has recorded historically large budget surpluses since 2022, driven overwhelmingly by concentrated corporate tax receipts from a small number of large multinationals — corporate tax receipts reached €23.8 billion in 2023, and the 2024 surplus reached roughly 7% of modified GNI, boosted further by a €13 billion Apple back-tax payment following an EU Court of Justice ruling. To its credit, the government has explicitly avoided simply spending this windfall pro-cyclically, instead channelling it into two new sovereign wealth vehicles (the Future Ireland Fund and the Infrastructure, Climate and Nature Fund) targeting combined assets of over €100 billion by the mid-2030s — showing at least a partial, hard-won lesson learned from the concentrated, transaction-tax-dependent revenue base that left the exchequer so exposed in the 2000s. That said, the underlying revenue base remains heavily concentrated in a handful of large taxpayers, leaving genuine ongoing volatility risk the government itself openly acknowledges.

Monetary policy since 2022: another rate mismatch, in the opposite direction. The ECB's aggressive rate hikes from 2022 to fight eurozone-wide inflation were arguably a tighter setting than Ireland's own inflation and growth conditions alone would have called for, again illustrating the structural cost of ceding monetary policy entirely to a currency union average.

A strong answer makes the throughline explicit: Ireland's fiscal policy has swung between pro-cyclical (2000s boom, arguably worsening the eventual crash) and genuinely countercyclical (COVID-19 response, and more recent windfall saving via the sovereign wealth funds), while its monetary policy has been entirely determined by the ECB throughout, for better (crisis-era low rates, QE) and worse (a policy rate that has periodically been too loose or too tight for Ireland's own specific conditions) — the central, recurring critical point being that fiscal policy is the only lever Ireland itself actually controls.

How the 30% Is Likely Allocated

Tier	Marks	What separates this tier
24-30%	24-30%	Covers multiple distinct periods (Celtic Tiger boom, financial crisis/bailout, recovery, COVID-19, recent surplus) with specific facts for each, explicitly evaluates fiscal policy as pro-cyclical vs. countercyclical at different points, correctly notes Ireland's monetary policy is entirely determined by the ECB, and reaches a genuinely critical (not merely descriptive) overall assessment.
15-23%	15-23%	Covers two or three periods with reasonable factual detail but with limited critical evaluation, or doesn't clearly address the ECB/monetary policy dimension.
6-14%	6-14%	General, largely undated description of "austerity" or "recession" without specific periods, figures, or critical evaluation.
0-5%	0-5%	No meaningful attempt.

Question B2 Total: 100% (60% of the Paper)